

A Parity Classification for the Vanishing-Moment Rademacher-Type Inequality

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Status. Candidate full solution, likely valid. The type-necessity question in Remark 3.18 is answered completely: odd tensor orders characterize type, whereas even orders are vacuous over the real scalar field.

1 Source question

Kirchner and Schwab [1] prove in Corollary 3.17 that Rademacher type p implies their estimate (3.18) for independent random variables with vanishing k th injective moment. In Remark 3.18, pages 14–15, they ask whether type p is necessary. The source statement spans the following two page crops.

$$\left\| \sum_{j=1}^M \otimes^k \eta_j \right\|_{L_q(\Omega; \otimes_{\varepsilon_s}^{k,s} E)} \leq \left\| \sum_{j=1}^M \eta_j \right\|_{L_q(\Omega; \otimes_{\varepsilon_s}^{k,s} E)} \leq 2(2kK_{q,p}\tau_p(E) + B_q) M^{-(1-\frac{1}{p})} \|\xi_1\|_{L_{kq}(\Omega; E)}^k,$$

which along with the definition (3.14) of $C_{q,p,k}^{\text{SL}}$ completes the proof of (3.13). $\square$

The estimate (3.16) of term (A) in the proof of Theorem 3.16 reveals the following analogue of Proposition 3.14 for independent (not necessarily identically distributed) random variables $\eta_1, \dots, \eta_M \in L_{kq}(\Omega; E)$ with vanishing k th moment.

Corollary 3.17. *Assume that $(E, \|\cdot\|_E)$ is of Rademacher type $p \in [1, 2]$. Let $q \in [p, \infty)$, $k, M \in \mathbb{N}$ and $\eta_1, \dots, \eta_M \in L_{kq}(\Omega; E)$ be independent E -valued random variables with vanishing k th moment, $\mathbb{M}_{\varepsilon}^k[\eta_j] = 0$ for all $1 \leq j \leq M$. Then,*

$$\left\| \sum_{j=1}^M \otimes^k \eta_j \right\|_{L_q(\Omega; \otimes_{\varepsilon_s}^{k,s} E)} \leq 4kK_{q,p}\tau_p(E) \left(\sum_{j=1}^M \|\eta_j\|_{L_{kq}(\Omega; E)}^{kp} \right)^{1/p}. \quad (3.18)$$

Remark 3.18. Optimality of the convergence rate $1 - \frac{1}{p}$ in (3.13) is ultimately related to the question whether it is *necessary* that the Banach space E has Rademacher type $p \in [1, 2]$ for (3.18) to hold for all finite sequences $\eta_1, \dots, \eta_M \in L_{kq}(\Omega; E)$ of independent E -valued random variables with vanishing k th moment.

Figure 1: Corollary 3.17 and the beginning of Remark 3.18 on source page 14.

For clarity, fix a real Banach space E , $1 \leq p \leq 2$, $q \geq p$, and $k \geq 2$. Write $S_{\varepsilon_s}^k E$ for the completed symmetric injective tensor power, whose norm on a symmetric tensor T is

$$\|T\|_{\varepsilon_s} = \sup_{f \in B_{E^*}} |\langle T, f^{\otimes k} \rangle|.$$

Let $\text{VM}_{k,p,q}(E)$ denote the existence of a constant $C < \infty$ such that every finite independent family $\eta_1, \dots, \eta_M \in L_{kq}(\Omega; E)$ satisfying $\mathbb{E}[\eta_j^{\otimes k}] = 0$ obeys

$$\left\| \sum_{j=1}^M \eta_j^{\otimes k} \right\|_{L_q(\Omega; S_{\varepsilon_s}^k E)} \leq C \left(\sum_{j=1}^M \|\eta_j\|_{L_{kq}(\Omega; E)}^{kp} \right)^{1/p}. \quad (1)$$

For the first moment, $k = 1$, it is evident that the choice $\eta_j := r_j x_j$ in (3.18), where $(r_j)_{j \in \mathbb{N}}$ is a Rademacher family on $(\Omega, \mathcal{A}, \mathbb{P})$ and $x_1, x_2, \dots \in E$, implies that the Banach space E has Rademacher type p . However, for higher-order moments, this question is more complex due to the injective tensor norm on the left-hand side. For odd orders $k \in \mathbb{N}$ and the space $E := \ell_1$ of summable real-valued sequences (which only has Rademacher type $p = 1$), the choice $\eta_j := r_j e_j$ shows that (3.18) cannot hold for any $p > 1$. Here, $(e_j)_{j \in \mathbb{N}}$ denote the standard unit vectors in ℓ_1 . In addition, the classical Khintchine inequalities (3.12) imply that, for any Banach space E , the convergence rate in (3.13) cannot be better than $1/2$ (one may take, e.g., $\xi_j := g_j x$, where $(g_j)_{j \in \mathbb{N}}$ is an orthogaussian family on $(\Omega, \mathcal{A}, \mathbb{P})$ and $x \neq 0$ is a non-zero vector in E). Sharpness of the rate $1 - \frac{1}{p}$ in (3.13) and necessity of the Rademacher type p for (3.18) for the case $p \in (1, 2)$ remains an open question.

The next lemma complements Theorem 3.16 when deriving convergence rates of single-level Monte Carlo methods for approximating injective k th moments.

Lemma 3.19. *Let $k \in \mathbb{N}$ and suppose that $\eta, \xi \in L_k(\Omega; E)$. Then,*

$$\|\eta \otimes \xi\|_{L_k(\Omega; E \otimes E)} \leq \|\eta\|_{L_k(\Omega; E)} \|\xi\|_{L_k(\Omega; E)} \quad \text{if } k \text{ is odd,} \\ \|\eta \otimes \xi\|_{L_k(\Omega; E \otimes E)} \leq \|\eta\|_{L_k(\Omega; E)} \|\xi\|_{L_k(\Omega; E)} \quad \text{if } k \text{ is even, } k \geq 2.$$

Figure 2: The continuation and open-question sentence on source page 15.

2 Proof intuition

Parity changes the question completely. At even order, every repeated dual evaluation of a zero tensor moment is the expectation of a nonnegative even power. Thus the random variable itself must vanish, so the estimate tests no geometry at all.

At odd order, Rademacher signs turn any deterministic vector z into an admissible random variable rz . Pure powers still hide the linear Rademacher sum. To recover it, split off a fixed transverse direction y_0 from a hyperplane H . For $x \in H$, balance $y = \|x\|^{1/k} y_0$ and $u = \|x\|^{-(k-1)/k} x$. Polynomial interpolation extracts the term linear in u from $(y + tu)^{\otimes k}$. The balancing ensures that its kp -cost is exactly $\|x\|^p$, while a functional equal to one on y_0 recovers the ordinary Rademacher sum in H .

3 Classification theorem

Theorem 1 (Parity classification). *Let E be a real Banach space, $1 \leq p \leq 2$, $q \geq p$, and $k \geq 2$.*

1. *If k is odd, then $\text{VM}_{k,p,q}(E)$ holds if and only if E has Rademacher type p .*
2. *If k is even, then every E -valued random variable with vanishing k th injective moment is zero almost surely. Consequently $\text{VM}_{k,p,q}(E)$ holds vacuously for every real Banach space E . In particular, for every $p > 1$, $E = \ell_1$ disproves necessity of type p .*

Even orders

Proof of Theorem 1(2). Let k be even and let $\eta \in L_{kq}(\Omega; E)$ satisfy $\mathbb{E}[\eta^{\otimes k}] = 0$. Strong measurability gives a separable closed subspace $E_0 \subseteq E$ containing $\eta(\omega)$ almost surely. There is a sequence $(f_n) \subseteq B_{E^*}$ which norms E_0 :

$$\|x\| = \sup_n |f_n(x)|, \quad x \in E_0.$$

Indeed, choose a dense sequence in the unit sphere of E_0 , take norming functionals there, and extend them to E by Hahn–Banach.

Applying $f_n^{\otimes k}$ to the Bochner integral gives

$$0 = \left\langle \mathbb{E}[\eta^{\otimes k}], f_n^{\otimes k} \right\rangle = \mathbb{E}[f_n(\eta)^k].$$

The integrand is nonnegative because k is even, so $f_n(\eta) = 0$ almost surely. Intersecting the countably many full-measure events and using the norming property gives $\eta = 0$ almost surely. Hence every family admissible in (1) is identically zero. The final assertion follows because ℓ_1 has no Rademacher type $p > 1$. $\square$

Odd orders: coefficient extraction

For $0 \leq \ell \leq k$, put

$$c_0 = -\sum_{m=1}^k \frac{1}{m}, \quad c_\ell = (-1)^{\ell-1} \frac{1}{\ell} \binom{k}{\ell} \quad (1 \leq \ell \leq k). \quad (2)$$

Lagrange interpolation at $0, 1, \dots, k$ implies that for every polynomial P of degree at most k , its coefficient of t is

$$[t]P(t) = \sum_{\ell=0}^k c_\ell P(\ell). \quad (3)$$

The formula follows by differentiating the Lagrange basis at zero. In particular, in the symmetric algebra over E ,

$$k \operatorname{sym}(y^{\otimes(k-1)} \otimes u) = \sum_{\ell=0}^k c_\ell (y + \ell u)^{\otimes k}. \quad (4)$$

Proof of Theorem 1(1). If E has type p , property $\mathbf{VM}_{k,p,q}(E)$ is precisely the sufficiency estimate proved in Corollary 3.17 of [1]. We prove the converse.

Assume (1) with constant C and let k be odd. Fix $y_0 \in S_E$ and $\psi \in S_{E^*}$ with $\psi(y_0) = 1$. Set

$$H = \ker \psi, \quad Px = x - \psi(x)y_0.$$

Then P projects E onto H and $\|P\| \leq 2$.

Take arbitrary $x_1, \dots, x_M \in H$. If $x_j \neq 0$, define

$$a_j = \|x_j\|^{1/k}, \quad y_j = a_j y_0, \quad u_j = a_j^{-(k-1)} x_j;$$

if $x_j = 0$, set $a_j = y_j = u_j = 0$. Thus

$$\|y_j + \ell u_j\| \leq (1 + \ell)a_j, \quad a_j^{kp} = \|x_j\|^p. \quad (5)$$

Let $r_1, \dots, r_M$ be independent Rademacher signs. For each fixed ℓ , the random variables $\eta_{j,\ell} = r_j(y_j + \ell u_j)$ are independent and, since k is odd,

$$\mathbb{E}[\eta_{j,\ell}^{\otimes k}] = \mathbb{E}[r_j^k](y_j + \ell u_j)^{\otimes k} = 0.$$

Applying (1) and (5) gives

$$\left\| \sum_j r_j (y_j + \ell u_j)^{\otimes k} \right\|_{L_q(S_{\varepsilon_s}^k E)} \leq C(1 + \ell)^k \left(\sum_j \|x_j\|^p \right)^{1/p}. \quad (6)$$

Put $T_j = \text{sym}(y_j^{\otimes(k-1)} \otimes u_j)$ and $D_k = \sum_{\ell=0}^k |c_\ell|(1+\ell)^k$. Summing (4) with signs and using (6) yields

$$\left\| \sum_j r_j T_j \right\|_{L_q(S_{\varepsilon_s}^k E)} \leq \frac{CD_k}{k} \left(\sum_j \|x_j\|^p \right)^{1/p}. \quad (7)$$

We now recover the ordinary Rademacher sum. For each realization, let $h = \sum_j r_j x_j \in H$ and choose $\phi \in B_{H^*}$ norming h up to an arbitrarily small error. Define

$$f(x) = \phi(Px) + \psi(x), \quad x \in E.$$

Then $f(y_0) = 1$, $f|_H = \phi$, and $\|f\| \leq \|P\| + \|\psi\| \leq 3$. Testing against $(f/3)^{\otimes k}$ and using the balanced definitions gives

$$\left\| \sum_j r_j T_j \right\|_{\varepsilon_s} \geq 3^{-k} |\phi(h)|.$$

Letting the norming error tend to zero, taking L_q -norms, and combining with (7), we obtain

$$\left\| \sum_j r_j x_j \right\|_{L_q(E)} \leq \frac{3^k CD_k}{k} \left(\sum_j \|x_j\|^p \right)^{1/p}. \quad (8)$$

Since $q \geq p$, the same bound holds in L_p . Thus H has Rademacher type p .

Finally, every $x \in E$ decomposes as $x = Px + \psi(x)y_0$. For arbitrary $x_1, \dots, x_M \in E$, the triangle inequality, (8), and the scalar Khintchine inequality give

$$\left\| \sum_j r_j x_j \right\|_{L_p(E)} \leq 2 \frac{3^k CD_k}{k} \left(\sum_j \|x_j\|^p \right)^{1/p} + \left(\sum_j |\psi(x_j)|^p \right)^{1/p}.$$

Therefore E has Rademacher type p . □

4 Consequences, verification, and scope

Theorem 1 supplies a complete answer to the necessity question for (3.18): affirmative at odd orders, negative and vacuous at even orders. The source paper's separate question about sharpness of the i.i.d. Monte Carlo rate (3.13) is not claimed here.

The exact script `code/verify_interpolation.py` checked (3) for every monomial of degree at most k and all $1 \leq k \leq 20$. This check is not used as a substitute for the displayed Lagrange proof.

A bounded literature search through 9 August 2026 covered the run indexes, the current arXiv and published versions of [1], exact searches using the phrases “vanishing k th moment,” “Rademacher type,” and “injective tensor,” and the related follow-up arXiv:2605.24620. No source recording the parity classification or the balanced-polarization proof was found. This is not a novelty guarantee.

Human-review recommendation. Check the normalization in (4), the balanced exponent in (5), and the pointwise choice of the norming functional before accepting the result as a full solution of the necessity question.

References

- [1] K. Kirchner and C. Schwab, *Monte Carlo convergence rates for k th moments in Banach spaces*, J. Funct. Anal. **286** (2024), Article 110218; arXiv:2212.03797.